INFO-H-502 –3DGRAPHICS
PROJECT GUIDELINES

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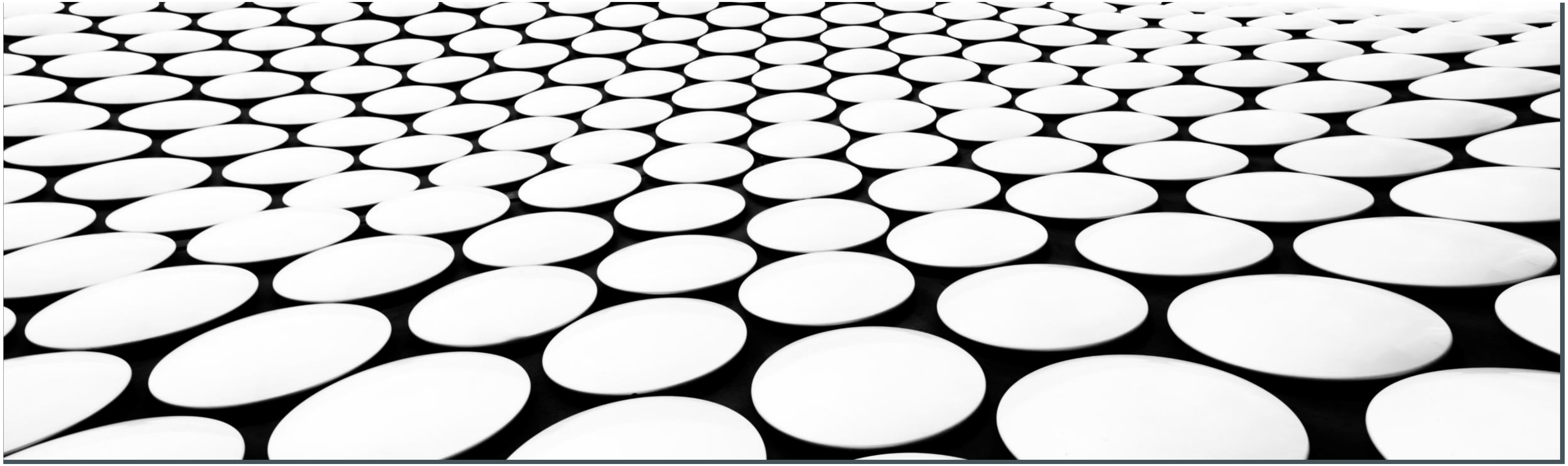

PROJECT INFO – MINIMAL REQUIEREMENT

Minimal requirements : To get a passing grade, the project should contain **ALL** basic features we saw in the exercise sessions and **AT LEAST ONE** of the intermediate feature seen during Exercice Sessions 7 or 8 (LAB04)
You **must** also explain how you would implement an additional feature, see slide 4

- Basic features:
  - lights (ambient, specular, diffuse, ..) (Session 5 - LAB 03)
  - Textures (Session 4 – LAB 02)
  - Loading (more than one) model (Session 5 - LAB 03)
  - Cubemap (Session 6 - LAB 03)
  - Game logic (Bachelor)
  - Object moving in the scene (Session 4 - LAB02)
  - free navigation within it (camera control) (Session 5 - LAB03)
  - Reflection/refraction (Session 6 -LAB03)
- Intermediate features:
  - Frame buffer object effect (diffusion shader, mirror and co)
  - Particles
  - Instancing
  - Billboarding

PROJECT INFO – ADDITIONAL FEATURES

- Implement some more advanced feature to get a higher grade, features will be graded according to their difficulty
- Some Advanced features:
  - Bump mapping
  - Collision detection + Physics (done by yourself or using libraries, bullet in C++ , ammo.js in javascript)
  - Shadows
  - Skeleton animation (with or without Skinning) (http://ogldev.org/)
  - 3D mesh read by a file by hand (Collada, gltf, ect.. ...)
  - Geometry shader
  - Compute shader
  - Tessellation
  - Toon shader
  - + Other ....

PROJECT INFO – LINK WITH THE THEORY

- Per group, choose one of the chapter listed from the three books below. You must choose a feature described in the chapter, try to implement it in your project and explain it in more detail during your project presentation. You will be graded on your explanation of the concept more than the result of the implementation. If you choose the same chapter as another group, make sure you pick different features

- “Real-time Rendering” by T. Akenin-Möller *et al.*
  - Chapter 7 : Shadows
  - Chapter 9 : Physically Based Shading
  - Chapter 10 : Local Illumination (ex : area light sources)
  - Chapter 12 : Image-Space Effects
  - Chapter 13 : Beyond Polygons
  - Chapter 14 : Volumetric and Translucency Rendering
  - Chapter 15 : Non-Photorealistic Rendering
  - Chapter 17 : Curves and Curved Surface
  - Chapter 22 : Intersection Test Methods
- "Texturing and Modeling : A Procedural Approach" by D. Ebert *et al.*
  - Chapter 4 : Cellular texturing
  - Chapter 11 : Procedural Synthesis of Geometry
  - Chapter 16 : Procedural Fractal Terrain
  - Chapter 18 : Atmospheric Model
  - Chapter 19 : Genetic Textures
- Non-Photorealistic Computer Graphics : Modeling, Rendering, and Animation by T. Strothotte and S. Schlechtweg
  - Chapter 4 : Simulating Natural Media and Artistic Technique
  - Chapter 8 : Lighting Model for NPR
  - Chapter 9 : Distorting Non-Realistic Rendition

INSPIRATION

- Some idea for scene you can implement :
  - Bowling/flipper games
  - Roller coaster
  - Skying game
  - Beach scene with realistic water simulation
- If you want to create a specific simulation which doesn't fit with the minimum requirements (ex : advanced fluid simulation but no textures) contact us so we can define specific requirement for your project.
- Need inspiration ? → You can find videos of past years here:
  - https://youtu.be/7tlVZoH4qzI
  - https://www.youtube.com/watch?v=yHYgNVRyfOA&list=PLjrvohwUr7rPKjgZeXOKODvqT9B6F3ygZ&index=5

SCHEDULE & DELIVRABLE

- Group of maximum 2 students, grade adjusted for the number of student
- You will submit on the UV :
  - A pdf report (around 5 pages, maximum 10 pages) with a brief description of your project and the features implemented, your report should include a link to a git repository with the code of your project
  - A screen capture video (mp4 or avi format) of the animation/scene in your project, showing the main features
  - The code for your project, in a zip file
  - A readme with library dependency if needed and instruction how to launch the project
- All your deliverable should be named LAST_NAME_STUDENT1_LAST_NAME_STUDENT2_2026.[pdf/mp4/zip]
- The project will be due in end of May
- **Don’t underestimate the amount of work needed ! You will need around 1-2 full weeks of work to get decent results !**